

Forecasting Canyon Lake Storage: A Multi-Model Approach to Supporting Regional Water Supply Agreements in Central Texas

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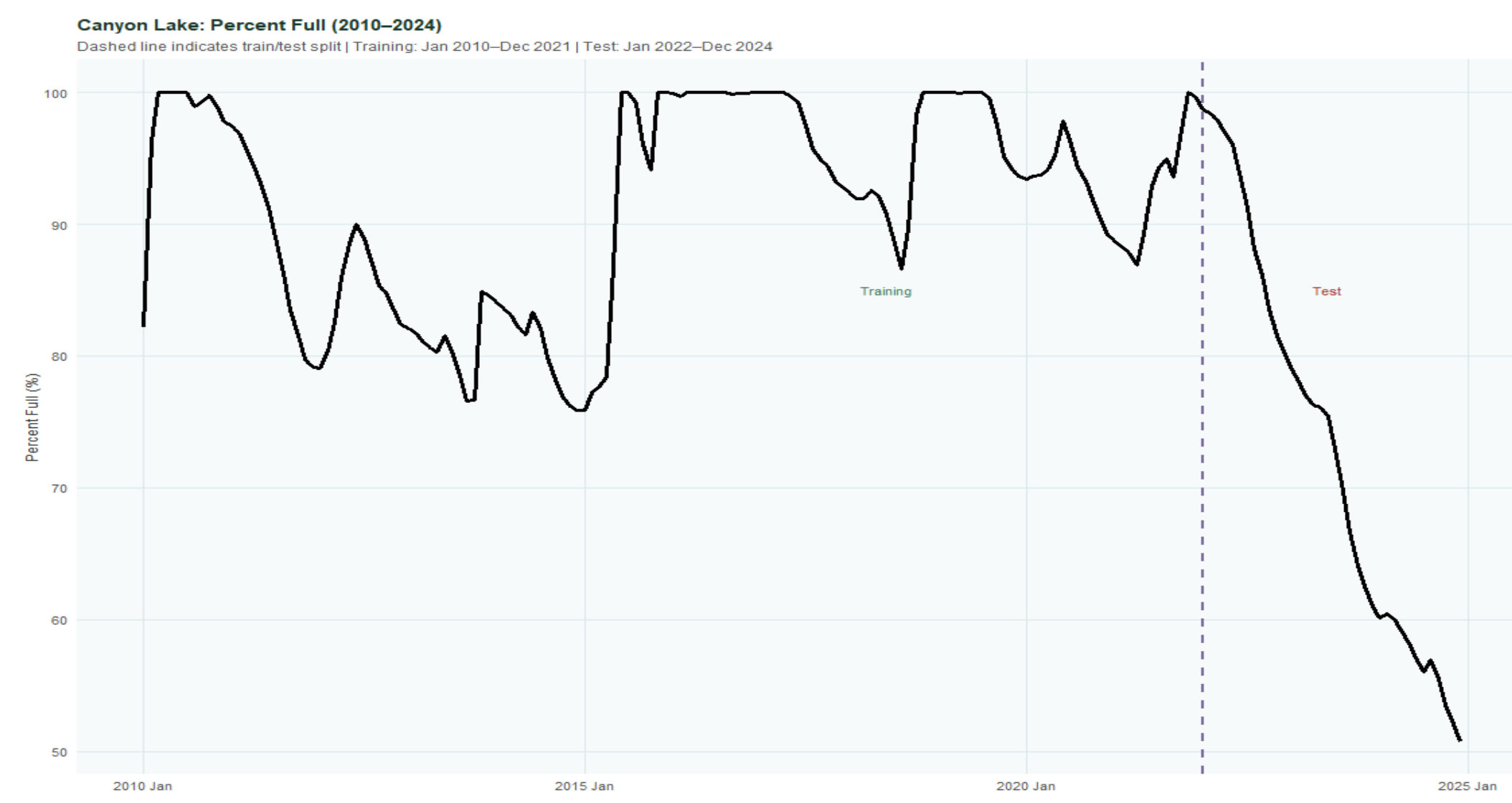
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Abstract

Water resource forecasting has become a critical planning tool for rapidly growing municipalities in Central Texas. As cities like Kyle, Texas pursue aggressive commercial and residential expansion, regional water supply agreements with neighboring municipalities depend on reliable long-term storage forecasts. This project constructs and evaluates four forecasting models — Exponential Smoothing (ETS), ARIMA, Time Series Linear Regression (TSLM), and an Ensemble average — using an 80/20 train/test split across 180 monthly observations. Models were evaluated on RMSE, MAE, and MAPE against a 36-month holdout period. While TSLM produced the lowest raw error metrics, residual diagnostics revealed significant autocorrelation in its errors, undermining the reliability of its prediction intervals. ARIMA emerged as the most defensible model, combining competitive forecast accuracy (RMSE = 30.4) with clean residuals confirmed by a Ljung-Box test ($p = 0.688$). For water utility planners and municipal decision-makers, ARIMA's ability to produce statistically valid confidence intervals makes it the most actionable tool for managing supply commitments under conditions of hydrological uncertainty.

Introduction & Significance

Canyon Lake, located in Comal County, Texas, serves as a primary surface water source for communities across the greater Central Texas region. The Guadalupe-Blanco River Authority (GBRA) administers the water for wholesale, and the reservoir's conservation levels directly govern water availability for municipal supply contracts throughout the Guadalupe River Basin. The city of Kyle currently receives 54% of its total permitted water supply from Canyon Reservoir, but contractually wholesale customers are subject to curtailments across six drought response stages, each triggered by declining reservoir levels. The reservoir has a total capacity of 123 billion gallons, but its current level is significantly reduced due to ongoing drought conditions, leading to water usage limits (City of Kyle, 2025). The city of Kyle has experienced significant growth over the last 25 years, increasing from 5,000 residents in 2000 to 50,000 in 2020, with projections reaching 150,000 by 2080. Kyle remained under Stage 3 drought restrictions through 2025, and as the city pursues continued economic expansion, accurate long-term water supply forecasting has become an operational necessity (KXAN Austin, 2025).



Time series forecasting is a well-established quantitative tool for resource planning, with applications ranging from energy demand to reservoir storage management. Das et al. (2023) demonstrated that time series models applied to Texas reservoirs provide water planners improved ability to make informed decisions about water usage, minimizing impacts to local communities and businesses. This project applies four forecasting methodologies — Exponential Smoothing (ETS), ARIMA, Time Series Linear Regression (TSLM), and an Ensemble average — to monthly Canyon Lake storage data spanning January 2010 through December 2024, following the modeling framework described in Hyndman & Athanasopoulos (2021). The goal is to identify the model best suited for supporting water supply planning decisions in a rapidly urbanizing region under recurring drought pressure.

Methods

Monthly percent-full observations for Canyon Lake were obtained from the Texas Water Data portal (waterdatafortexas.org) covering January 2010 through December 2024, yielding 180 observations. Daily observations were aggregated to monthly averages and structured as a tsibble object in R using the fpp3 and tidyverse packages. The dataset was divided into a training set (January 2010 – December 2021, $n = 144$) and a test set (January 2022 – December 2024, $n = 36$), reflecting an approximate 80/20 split. All model fitting, transformation assessment, and stationarity testing were performed exclusively on the training data. A Box-Cox transformation was assessed using the Guerrero method but was not applied, as the bounded nature of the percent-full variable (0–100%) limits the appropriateness of variance-stabilizing transformations. Seasonal differencing and unit root tests (KPSS) were used to assess stationarity prior to ARIMA specification.

Four models were fit to the training data:

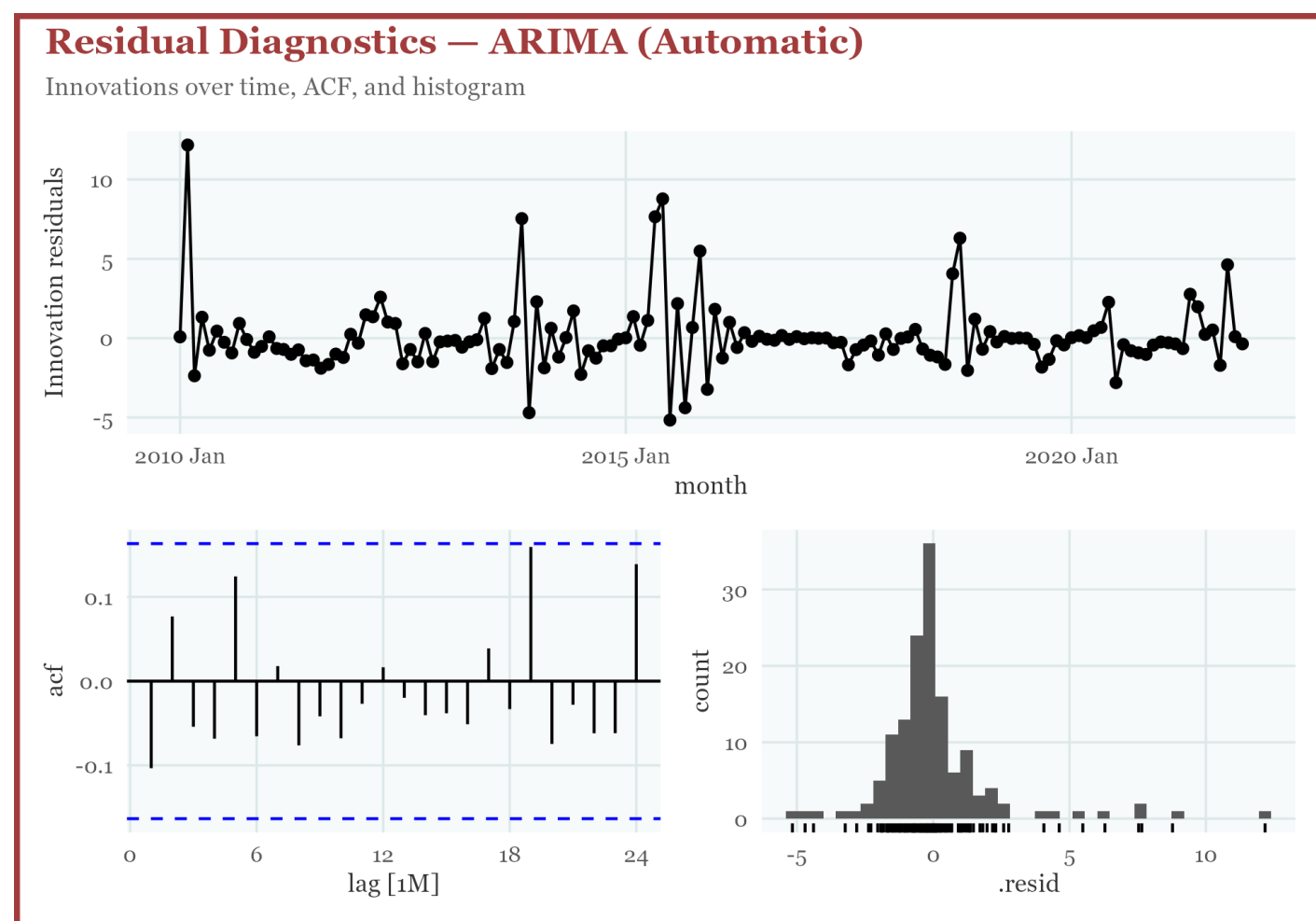
- Exponential Smoothing (ETS)** was selected automatically, and weights recent observations more heavily than older ones.
- ARIMA** was selected automatically, and captures the relationship between current and past values, which is common in reservoir storage series.
- Time Series Linear Regression (TSLM)** was fit with both trend and seasonal dummy terms, modeling storage as a linear function of time and month of year.
- Ensemble** was constructed as a simple arithmetic average of the point forecasts from the above models, with the goal of reducing forecast variance by combining different modeling approaches.

Results

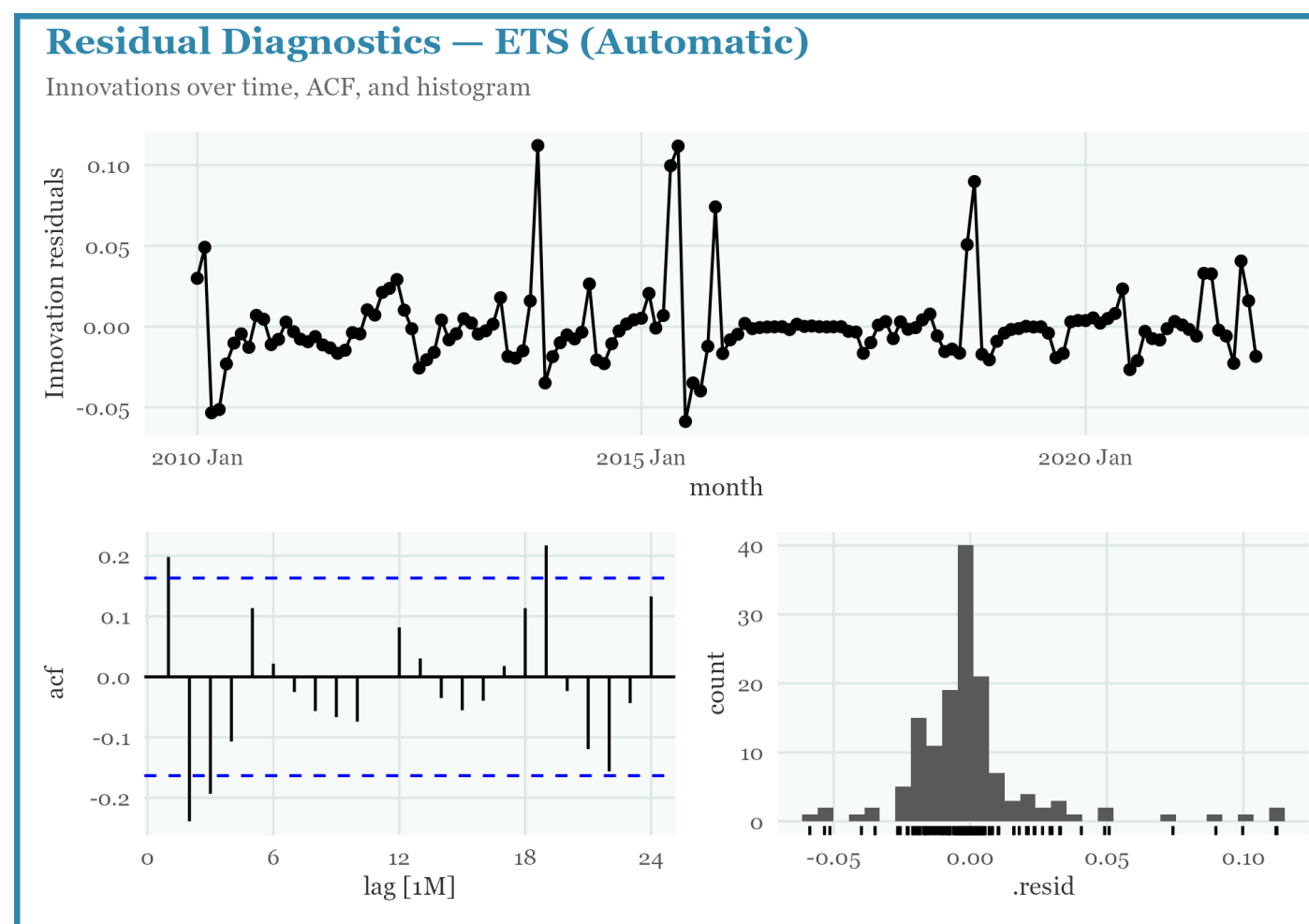
Table 1 summarizes forecast accuracy across all four models on the 36-month test set.

Model	RMSE	MAE	MAPE
TSLM	28.5	24.3	39.0%
ARIMA	30.4	26.4	42.0%
Ensemble	30.6	26.4	42.2%
ETS	33.0	29.1	46.0%

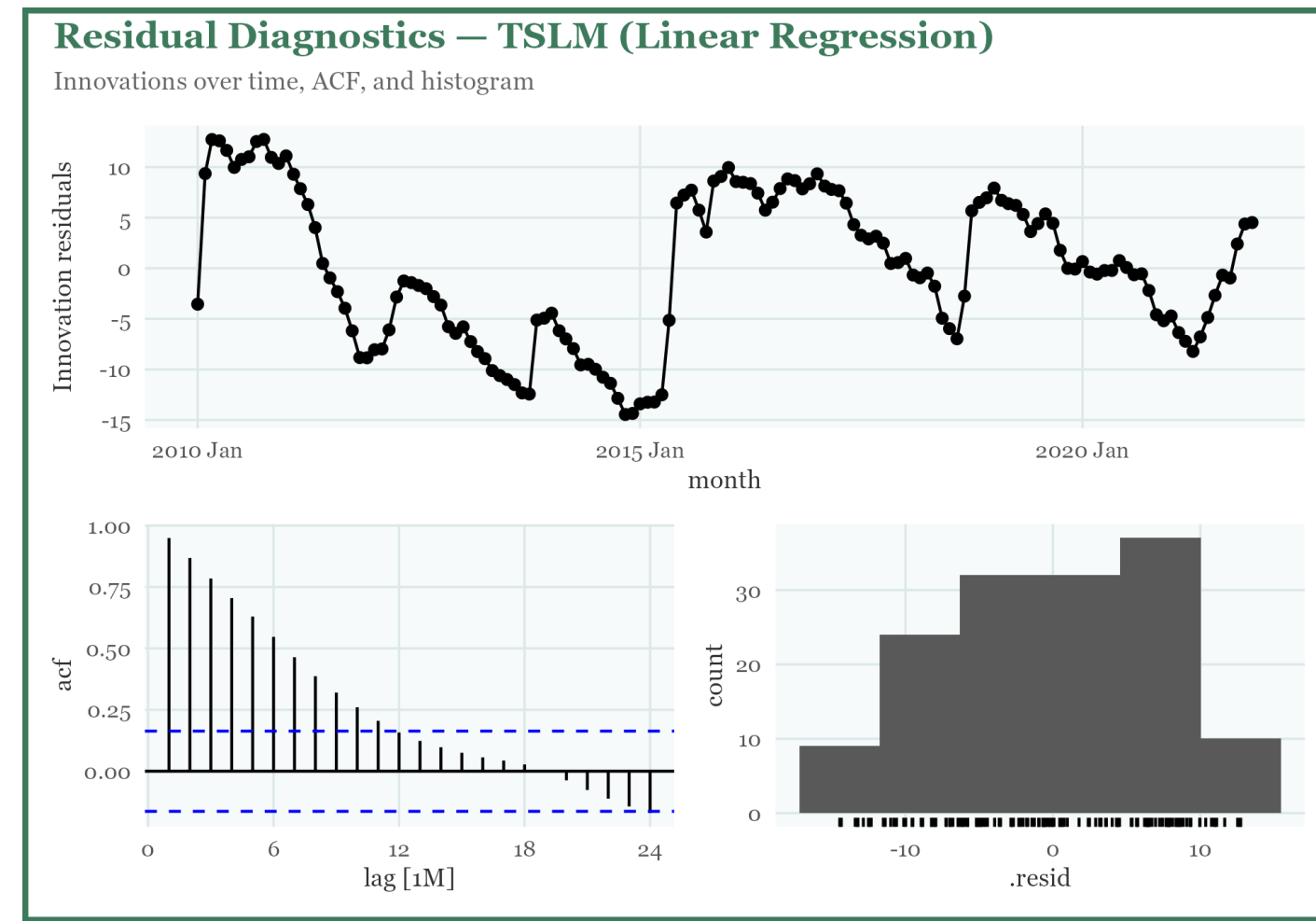
While TSLM recorded the lowest error on all three metrics, Ljung-Box testing revealed significant residual autocorrelation for both TSLM ($p \approx 0$) and ETS ($p = 0.002$), indicating that these models did not fully capture the dependence structure in the data. ARIMA was the only model to produce statistically clean residuals ($p = 0.688$), suggesting its error terms approximate white noise and that its prediction intervals are statistically valid.



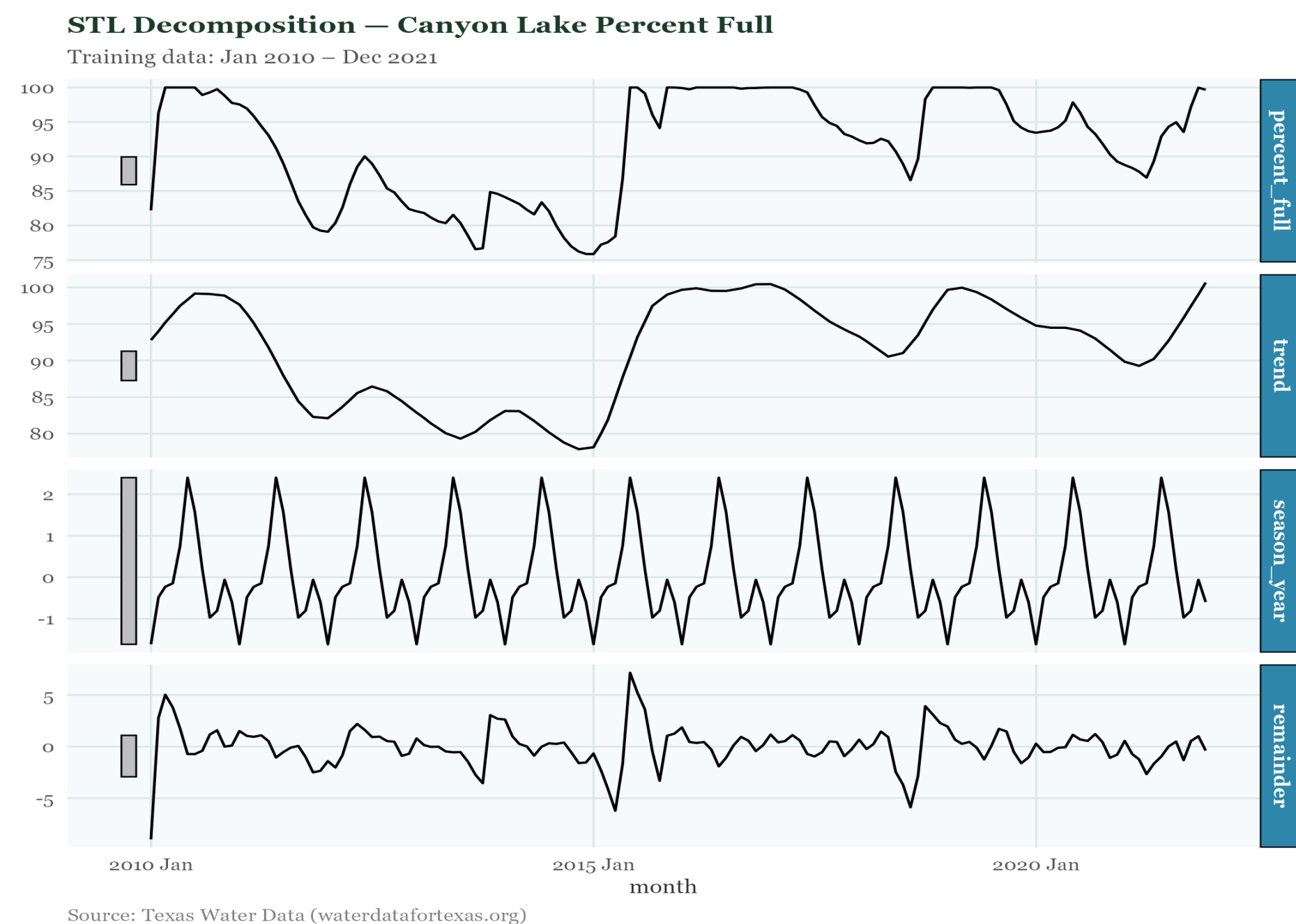
Residuals approximate white noise with no significant autocorrelation in the ACF, confirmed by a Ljung-Box test ($p = 0.688$), indicating the model adequately captured the series structure.



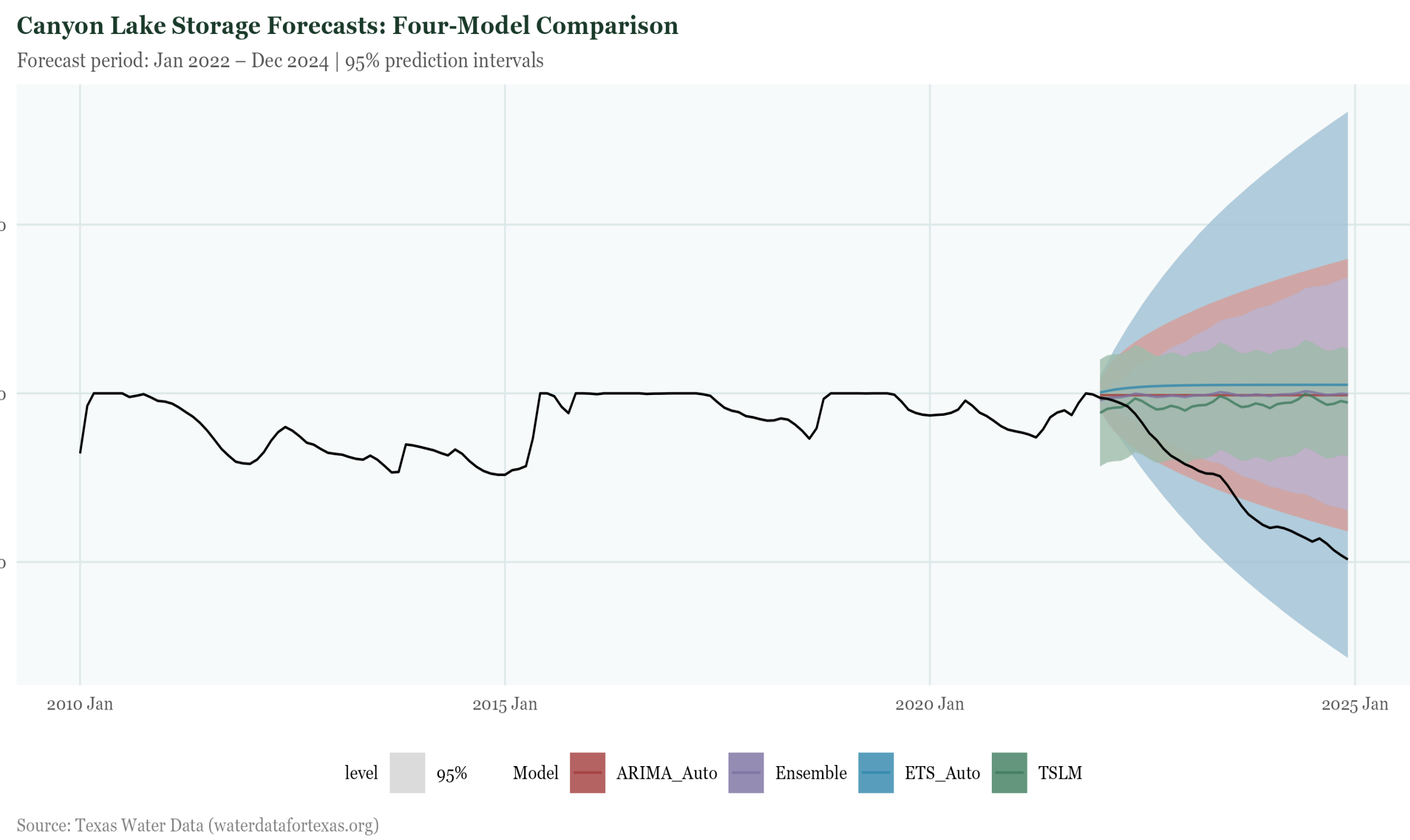
The ACF plot shows significant autocorrelation at multiple lags, confirmed by a Ljung-Box test ($p = 0.002$), indicating the model did not fully capture the dependence structure of the series.



Severe autocorrelation across all lags in the ACF, confirmed by a Ljung-Box test ($p \approx 0$), reflects a fundamental limitation of linear regression applied to correlated time series data. Point forecast accuracy is not a reliable indicator of model validity in this case.



Components shown top to bottom: observed series, trend, seasonal cycle, and remainder. The strong annual seasonal pattern reflects consistent summer drawdown and winter recovery, while the irregular remainder captures drought-driven departures from trend.



All four models generated 36-month ahead forecasts evaluated against the test set using RMSE, MAE, and MAPE. Residual adequacy for the three individual models was assessed using the Ljung-Box test at lag=24 and visual inspection of ACF plots.

Discussion

(explain usefulness of chosen models, limitations, and suggestions for improvement or future work)

Based on the combined evidence of forecast accuracy and residual diagnostics, the ARIMA model is recommended as the most suitable forecasting tool for Canyon Lake storage planning. Although TSLM produced marginally better point forecast accuracy, its severely autocorrelated residuals indicate a structural misspecification that invalidates its confidence intervals. In a water supply planning context, prediction intervals matter as much as point forecasts — a utility negotiating a multi-year supply agreement with a growing municipality like Kyle needs to know not just the expected storage level, but the realistic lower bound of that estimate under adverse conditions.

The ARIMA model satisfies both requirements: it performs competitively on accuracy while producing residuals consistent with white noise, meaning its uncertainty estimates can be trusted for contractual and operational decision-making.

Several limitations should be noted. First, the model is univariate and does not incorporate precipitation forecasts, upstream inflow data, or demand-side variables, all of which would meaningfully improve predictive power. Second, the high MAPE values across all models (39–46%) reflect the inherent difficulty of forecasting a climatically sensitive variable over a 36-month horizon — Canyon Lake levels are heavily influenced by drought cycles and episodic rainfall events that no purely historical model can anticipate. Future work could incorporate ENSO indices or NOAA precipitation outlooks as exogenous regressors within an ARIMAX or dynamic regression framework, potentially improving both accuracy and planning utility for regional water managers.

Conclusion

Canyon Lake storage forecasting is not a purely academic exercise — it directly informs the water supply agreements that Central Texas municipalities depend on as the region grows. This study demonstrates that model selection cannot rely on accuracy metrics alone. TSLM's lower RMSE is undermined by structurally invalid residuals, while ARIMA's clean diagnostics confirm its uncertainty estimates are trustworthy for operational planning. For water utilities operating under GBRA's drought curtailment framework, the lower bound of a forecast matters as much as the point estimate. ARIMA provides both. As Kyle and surrounding communities continue to expand, reliable reservoir forecasting will remain essential to sustainable regional water management — and a foundation for more sophisticated multivariate models incorporating precipitation, inflow, and demand-side data in future work.

References

- About Kyle Water | City of Kyle. (2022). City of Kyle. <https://www.cityofkyle.gov/city-services/water-utilities/about-kyle-water/>.
- Stark, S. (2025, May 14). City of Kyle diversifying water sources amid pervasive drought conditions. KXAN Austin. <https://www.kxan.com/news/local/hays/kyle/city-of-kyle-diversifying-water-sources-amid-pervasive-drought-conditions/>.
- Rohli, E., Woolsey, N., & Sathiaraj, D. (2023). Near-term forecasting of water reservoir storage capacities using long short-term memory. *Environmental Data Science*, 2, e30. doi:10.1017/eds.2023.25
- Drought Contingency Plan for Wholesale Customers. (n.d.). Retrieved March 20, 2026, from https://www.gbra.org/wp-content/uploads/2024/05/2024-Wholesale-Drought-Contingency-Plan_FINAL-w-Resolution.pdf
- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). In otexts.com. OTexts. <https://otexts.com/fpp3/>
- Water Data For Texas. (2026). Waterdatafortexas.org. <https://www.waterdatafortexas.org/reservoirs/individual/canyon>



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